



Tema: System Definition, Requirements Definition, and Conceptual Modeling

Um Caso para Estudo

The Community Library of Vila Verde has experienced significant growth over the past few years, both in the number of books available and in the number of users who rely on its services. With the increase in the collection and the volume of daily operations, it has become increasingly difficult to efficiently control which books are available, which are on loan, and which are reserved.

To improve organization and facilitate information management, the library's management decided to create a database that would allow all relevant information to be stored and quickly accessed. The main objective is to make the process of recording loans and reservations simpler, as well as to ensure better control over the history of operations.

Thus, a list of requirements was prepared with the main aspects to consider in building the database. In particular, it was defined that each book must be registered with a unique identifier, title, author, and category. The library also intends to record information about the authors of the books. Regarding users, the intention is to record an identifier, name, and other relevant details. The library also has staff members responsible for customer service and recording operations, making it necessary to register the identifier and name of each staff member. The organizational structure is hierarchical, so one staff member may supervise other staff members.

Whenever a book is loaned, the loan date must be recorded, identifying the user who borrowed the book and the staff member who registered the transaction. In addition, when a book is not available, it must be possible to make a reservation, recording the reservation date and the respective user.

With this database, the library aims to improve information organization, facilitate data retrieval, and make processes more efficient.

(...)

Based on the presented case study and considering the various steps of the database development methodology, you are asked to:

- a) Analyze the various aspects of the case study, justify and explain why a database system should be used.
- b) Develop a list of descriptive requirements that allows the creation of a conceptual schema for a database for the presented case.
- c) Using Chen notation, sketch a CONCEPTUAL schema for a database capable of accommodating the information revealed by the presented case. Identify and present several schema characterization tables that should be included in your data dictionary.